White Papers: RAMANATH SHANBHAG

[KEYNOTE SPEAKER] Guessing Quality? – SELENIUM DAY CONFERENCE, 2020

How can testers make an impact? - Can you bring predictability through automation and right leadership? - How to answer customer questions like show me the ROI? - What are the key ingredients for a successful delivery? So, Guessing Quality?, is an attempt to look at the "day in the life of a tester", and answer questions on career, choices of tools and technology, and share experiences on how test automation and situational leadership helped us gain the trust of the customer on some of the challenging projects and their turnaround stories.

[WHITEPAPER] Dynamic Test Engineering Platform: CI and TestOps: STARWEST, 2014, USA

In today's environment of shrinking cycle times, growing budget pressures, and increasing demand for application agility, it is important that the handshake between development, build & release management, testing and operations is seamless. We need to do this without compromising on quality, leveraging automation, and collaboration between teams. Now-a-days, Enterprises are moving from just doing "application testing" to "end-to-end release testing". Key takeaways:

- Integrated test methodology with MindTest -ISO29119 certified which improves governance, project management, accuracy of estimates and transparency.
- Zero Infrastructure: Tests runs using cloud infrastructure (on-demand)
- Automated Deployment and Tests –predictable quality at early stages
- Dynamic Code coverage at compilation ensuring quality of behaviors and helps in determining a quantitative measure of code coverage, which indirectly measure the quality of the application or product
- Tools Standardization to provide end-to-end coverage
- Guaranteed Levels of service

[WHITEPAPER] Achieving Testability in Performance Testing: Google Test Automation Conference (GTAC), 2011

(https://www.youtube.com/watch?v=SN6ezNAdkr8)

Performance testing is a niche area which requires a different mindset as compared to testing an application for its functional correctness.

Performance testing affects these 3 entities the most – the customer, the end users and the performance testing team. The performance testing team is the liaison between an end user's experience of the application and a customer's expectations. In this presentation we have tried to summarize the inherent challenges faced by a performance test team to fulfill the requirements of the other stakeholders. We also present our approach to overcome these challenges based on experience.

Though there are ample tools available (both commercial and open source) most of them are designed to solve a specific need. Because of this you need to put in more efforts to tailor the tools together and cater to the requirements of a different performance project. For ex: one project might have a java based application stack deployed on a linux platform whereas in another project you could have an asp.net application on a windows, the performance test team should have tools to monitor both the scenario.

Performance testing projects generally have a very short duration because of which there is a very high pressure on the delivery team. To give you an example many a times the customer approach us, just before the holiday season like Christmas and New Year, to ensure that their application can handle the holiday traffic. This gives the performance testing team very little time span to successfully plan and execute the project.

Even if you have the right set of tools and team who are ready to deliver under high pressure, it doesn't necessary imply that you'll get things right the first time. You can easily end with the wrong results altogether by making a single mistake which can cost you a lot of time to correct. Its lot like traveling by air, if the flight goes off by a few degrees then you might end up at a different location altogether.

Expertise draws the line between a smoothly executed project and a sloppily executed project. Having the right expertise is imperative for the successful execution of any PT projects. After all you need trained and experienced Pilots to keep correcting flight route in case of the aircraft goes off course.

In today's scenario, the quality of service provided is measured through customer satisfaction. Since customer requirements are tightly bound to an end user's expectations; it is imperative to ensure that a website has good end user experience thereby increasing the number of revisits to a website which finally translates to happy customers.

At MindTree, since we have been doing performance testing for the past 10+ years across various technologies, we have been able to identify the factors which bring testability to performance testing.
All the challenges mentioned above makes it compulsory to take a multi-pronged approach towards performance testing. This would include having frameworks to carry out end-to-end activities; developing the accelerators to give an additional impetus; a time tested methodology to avoid common pitfalls and the finding the key to build the right expertise.

Though each of these entities are powerful on their own, to attain the ultimate solution you should be able to combine and harness all these artifacts, namely the methodology, framework, expertise and accelerators to reduce the time and effort required at the same time increase the value.

At GTAC 2010, we will go in detail on how we achieved testability in performance testing by combining these powerful artifacts and creating a noticeable value proposition to our customers.

[WHITEPAPER] SLA based model for testing: STeP-IN SUMMIT 2010

Clients' confidence in our capabilities and to accelerate trust in our relationships with first-time customers. Giving a customer the comfort feeling early in the relationship, providing assurance about what they expect, and a general commitment on how we would be conducting business can be leveraged through SLAs.

Committing to SLAs is like gambling; wagering on how the skills and competencies of our testers will address customer needs. If we have an objective understanding of our testers' capabilities with respect to customer needs, we can consider taking the calculated risk of offering an SLA, and assuring the customer both that what we have committed is achievable and that we have a plan for cases when we may not meet the first-tier objective. The end goal of the SLAs is customer trust and agreement on the work to be done in greater detail than if only top level goals are specified. For example, using SLAs like number of test cases executed per person day (or resource day/time unit), number of defects found per person over unit time, number of test cases automated per resource time, defects found in test automation code, or test case effectiveness- defects found through test passes Vs. ad-hoc method can give a customer better visibility on the service offered.

Risk / reward balance is key. Good SLAs are specific, quantifiable (a known maximum exposure), and measurable. The best SLAs are all these and we internally are confident that we (not the customer) will be the first to notice failure to meet the agreement so we can address the issue proactively. It is also important to create a culture of awareness of SLAs in which all internal and external stakeholders and project participants understand the SLAs and report potential failure to achieve them.

In this presentation, we'll describe a method we use to create SLAs from before customer engagement, during the RFP process, and through project execution and completion.

[WHITEPAPER] Building an Automation Framework around Open-Source Technologies. TESTING LEADERSHIP AWARD: QAI STC 2009

Re-usable test automation frameworks coupled with open-source tools and technologies is a key solution to shrink test cycle times and related costs. This paper covers an approach for creating a script-less web application automation framework around SeleniumRC which is robust, flexible, and extensible. Our experimental results on this framework show increased productivity and ROI, reduced learning curve and dependency on skilled resources, eliminated usage of commercial tools and manual configuration. The framework provides ease of script generation & management, concurrent and remote execution, test data generation and customized reports with screenshots. Preliminary statistics indicate that the automation framework reduces ~ 40% of total test execution effort per release cycle under a given scenario.

Citation provided by QAI: *The paper clearly extends the concept of automation to open source technology making it relevant to small & medium scale enterprises, and/or as a test bed for R&D. This innovative and tested idea is awarded for its potential value to testing community.*

[WHITEPAPER] Are You Ready to Move to Outcome-Based Relationships with Your Testing Partner? QAI STC 2009

Outcomes are about achieving a certain milestone. That is, the customer pays only if an expected deliverable is achieved with a defined level of quality. The outcomes are defined in terms of price, deliverables, and service levels. They are built to help clients achieve enhanced savings, productivity, and value. In outcome-based engagements, the service provider is directly responsible for managing its resources, and not simply the level of effort. Risk-reward is the key. Here the service provider shares some level of risk with the customer. Success of such relationships lies in taking advantage of the service provider's technical competency, processes, and tools to save money, improve productivity, and achieve other, potentially more valuable benefits.

[WHITEPAPER] Methodical Approach to Creating a Test Automation Strategy: STEP-In 2008

(http://www.scribd.com/doc/3610838/Methodical-Approach-to-Creating-a-Test-Automation-Strategy)

In this paper we will discuss Mindtree's Test Automation Strategy Service which offers a complete solution to this challenge. We have a rich experience of executing product focused test automation, and have carried out Test Automation projects for a number of very demanding situations. Using this practical experience we have designed a systematic methodology for Test Automation. In the strategy phase, we understand the test organization, existing test practices, testing problems, and past successes. Next we help define test automation goals for critical areas (i.e. where automation is the only solution), and to obtain cost and time benefits wherever possible. We analyze the functional test coverage and propose, if applicable, additional automation test cases to improve the coverage. We then put together a plan to implement automation which would deliver the greatest ROI. The plan includes a well-defined test automation vision, answers questions such as what, when and how of automation, and proposes a prioritized test automation plan. The strategy will also include recommendations on test automation tools and harnesses to assist with the "how" question.

ARTICLE: Analysis - How successful is your test automation strategy?

Test automation is always viewed as the most desired approach for testing due to the benefits it offers – cost saving, time saving, and reliability. With product life cycles getting squeezed, and IT budgets crashing through the roofs, test automation has become a critical goal for technology organizations, and one that must be planned carefully, writes Ramanath Shanbhag, of MindTree Ltd.

With increasing budget pressures and expectations to deliver more in the same engineering time, today's managers are faced with situations where they have to make the right decisions quickly. However, most often than not, they are unsure of whether the decisions they have taken are correct, and will produce the desirable results until they are half way through the project. With project releases shrinking, it only adds to the effect.

See more at: https://www.electronicsweekly.com/?s=ramanath